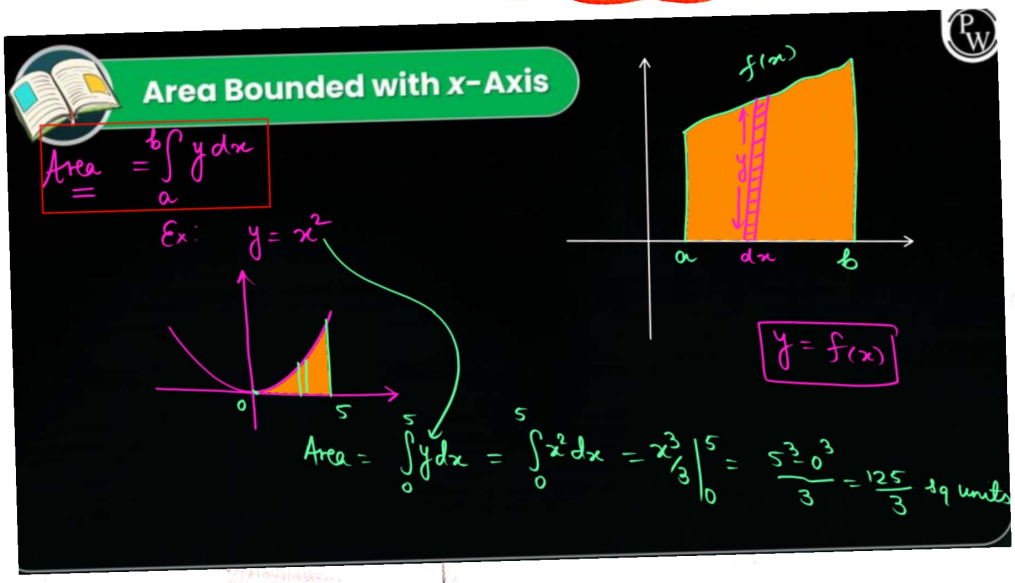
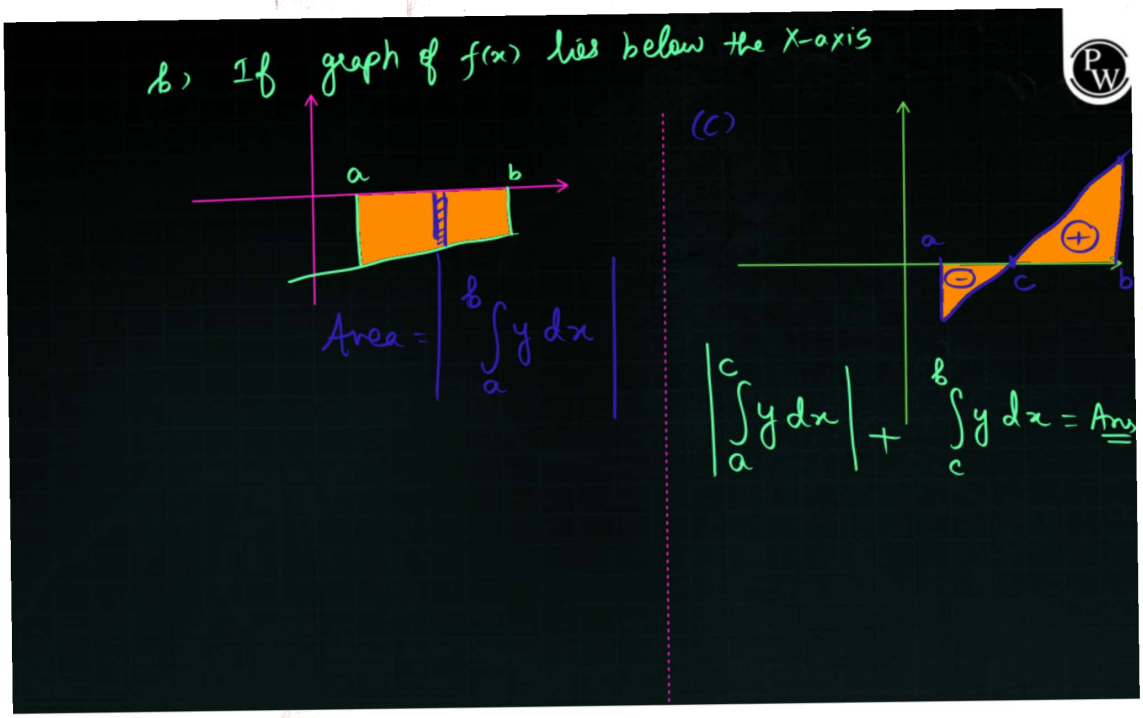
:- 12th BOARD HACKER JUGAADU NOTES :-

Area Bounded with x - Axis

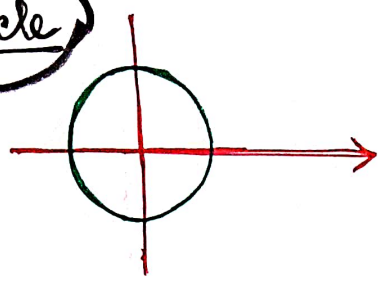


b, If graph of $f(x)$ lies below the x-axis



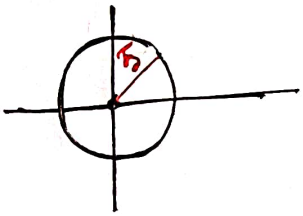
Some Common Graphs

Circle



$x^2 + y^2 = r^2$

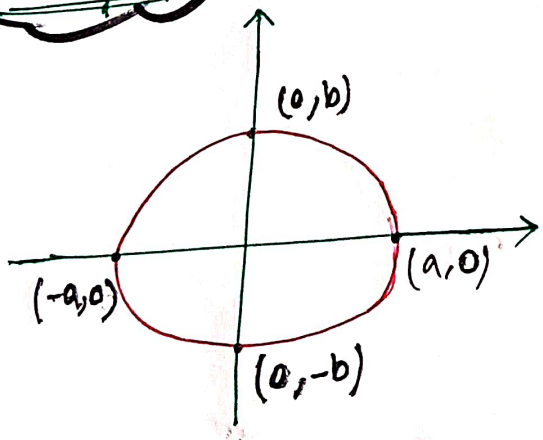
Centre $(0,0)$ & radius $= r$



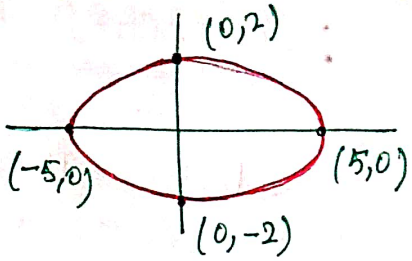
Ex:- $x^2 + y^2 = 25$

$r^2 = 25 \Rightarrow \boxed{r = 5}$

Ellipse



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

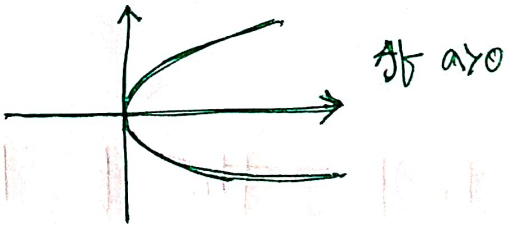


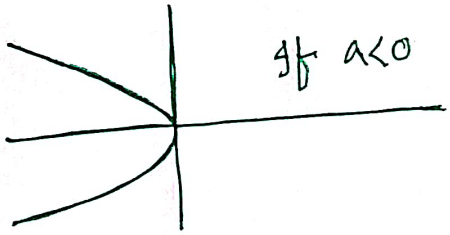
Ex:- $\frac{x^2}{25} + \frac{y^2}{4} = 1$

$\boxed{a = 5}$ & $\boxed{b = 2}$

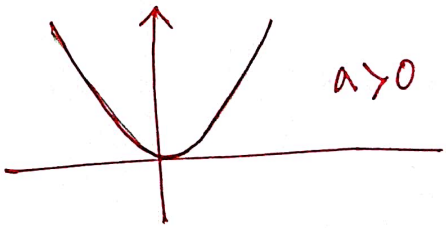
Parabola :-

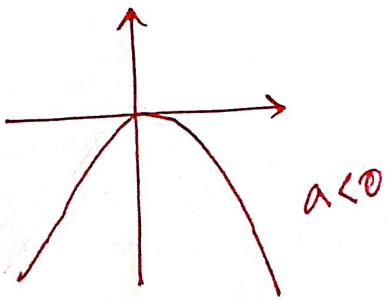
$y^2 = 4ax$



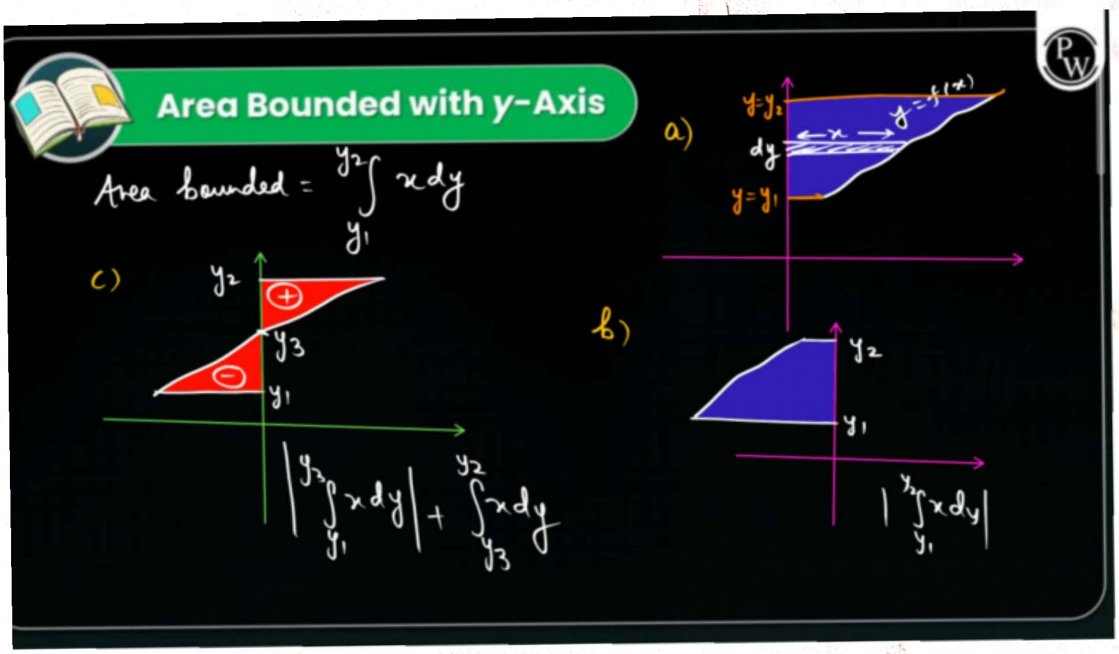


$x^2 = 4ay$





Area Bounded with y-Axis



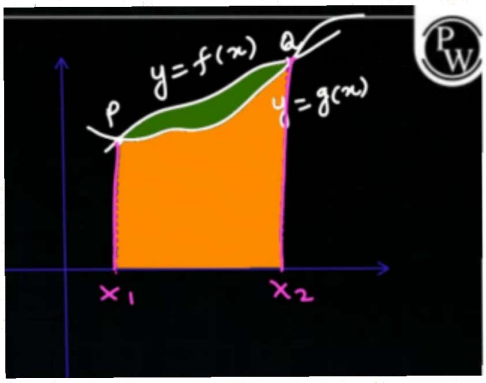
Area Bounded between 2 Curves using X-Axis

Get P.O.I of 2 Curves $\begin{cases} x_1 \checkmark \\ x_2 \end{cases}$

$$\text{Area bounded} = \int_{x_1}^{x_2} f(x)\,dx - \int_{x_1}^{x_2} g(x)\,dx$$

$$= \int_{x_1}^{x_2} (f(x) - g(x))\,dx$$

$$\text{Area} = \int_{x_1}^{x_2} (Y_{upper} - Y_{Lower})\,dx$$



Area Bounded between 2 Curves using Y-Axis

$$\int_{y_1}^{y_2} x_{Right}\, dy - \int_{y_1}^{y_2} x_{left}\, dy$$

Area bounded =

$$= \int_{y_1}^{y_2} (x_{Right} - x_{left})\, dy$$

